Nominal Concord needs Harmonic Serialism
Evidence from Archi

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1. Main Claim

- Concord in Archi ist best analysed as feature copying in Harmonic Serialism.
- An existing approach posits an unrestricted copy operation (Polinsky, 2016).
- A parallel contraint based approach (SPOT) faces the sour grapes problem (McCarthy, 2009) and cannot derive gapped structures.
- A Harmonic Serialism account avoids these problems by allowing opaque interaction of a local operation.

2. Data: Concord in Archi

- Gender concord in Archi (Nakh-Daghestanian, Russia) occurs on participles, numerals, demonstratives, and pronominal possessors.
- It never occurs on quantifiers, nominal possessors and nominal adjectives.

(1) čeħ hiba-tːu-**t** adam
not.one be.good-ATTR-IV.SG person(IV)[SG.ABS]
'not one good person' (Bond & Chumakina, 2016b, 116)

(2) to-**b** ɬːej<**b**>u doːˁzu-**b** χošon
that-III.SG five<III.SG> be.big-ATTR-III.SG dress(III)[SG.ABS]
'those five big dresses' (Bond & Chumakina, 2016a, 56)

(3) ɬːʷak-du-**t** maˁarul duχriqˁ
near-ATTR-IV Avar village(IV).LOC
'in the nearby Avar village' (Polinsky, 2016, 230)

- Similarly, low adverbs share the agreement of the verb.

(4) Patʼi ditːa<**r**>u da-qˁa
Pati(II).SG.ABS early<II.SG> II.SG-come.PFV
'Pati came early.' (Bond & Chumakina, 2016b, 113)

3. Assumptions: Serial Copying

- NP-over-DP structure with all modifiers as specifiers (Georgi & Müller, 2010)
- Harmonic Serialism (McCarthy, 2010)/Extremly Local Optimization (Heck & Müller, 2007) with a local feature copy operation and general feature delete operation (cf. Breteler (2017)).
- Feature coocurrence constraints (Gazdar et al., 1985) and LEFT(γ,XP) as a trigger of concord.

(5) Operations for nominal concord
a. COPY: A feature F with an index i on a head X is copied onto another adjacent head Y. This yields a feature F with an index i on head Y.
b. DELETE: A feature F with an index i on a head X is deleted from the head X. There is no feature F with an index i on head X anymore.

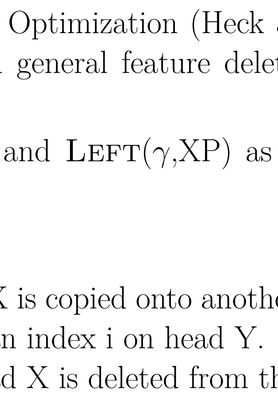
(6) Locality restriction on COPY
A head X is adjacent to a head Y iff
a. There is a node Z, such that
(i) Z directly dominates YP and
(ii) Z dominates X
b. and there is no head A, such that
(i) YP c-commands A and
(ii) AP c-commands X.

(7) a. *γ/X: Count one violation for every head that bears a categorical feature X and a gender feature.
b. LEFT(γ,XP): The highest head in XP should bear a gender feature.

5. Extensions: Adverbs and beyond

- The account of the nominal concord can be extended to adverbial agreement.
- Constraint against gender features on v and constraint requiring the highest head in the VP to bear gender features restrict agreement to lower adjectives.
- Reranking of the constraints MAX(γ) and *γ/N yields a Bantu-like noun class pattern.
  - Other possible rankings predict i.a. no concord or full condord.

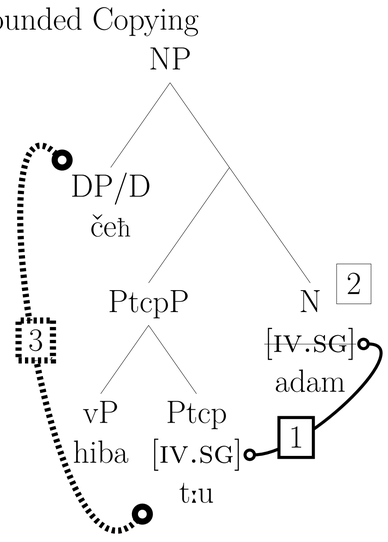
(21) Concord on adverbs



4. Analysis

- The local copy operation combined with serial evaluation allows for an iterative, bounded concord process, potentially including intermediate feature hosts.
- The high ranked *γ/D constraint bounds the copy operation inside the NP.

(8) Bounded Copying



(9) Derivation of (1) - 1

[NP PtcpP NIV]	*γ/D	LEFT(γ,NP)	*γ/N	MAX(γ)
a. [NP PtcpP NIV]		*!	*	
☞b. [NP PtcpPIV NIV]			*	
c. [NP PtcpP N]		*!		*

(10) Derivation of (1) - 2

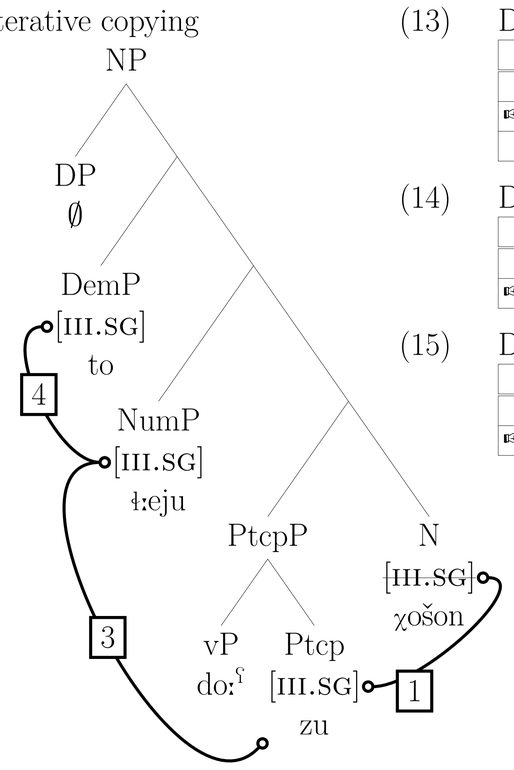
[NP PtcpPIV NIV]	*γ/D	LEFT(γ,NP)	*γ/N	MAX(γ)
a. [NP PtcpPIV NIV]			*!	
☞b. [NP PtcpPIV N]				*
c. [NP PtcpP NIV]		*!	*	*

(11) Derivation of (1) - 3

[NP DP [PtcpPIV N]]	*γ/D	LEFT(γ,NP)	*γ/N	MAX(γ)
☞a. [NP DP [PtcpPIV N]]		*		
b. [NP DP [PtcpP N]]		*		*!
c. [NP DPIV [PtcpPIV N]]	*!			
d. [NP DP [PtcpPIV NIV]]		*	*!	

- Merge adds a new highest head to the NP, allowing LEFT(γ,NP) to trigger copying iteratively.

(12) Iterative copying



(13) Derivation of (2) - 1

[NP PtcP NIII]	*γ/D	LEFT(γ,NP)	*γ/N	MAX(γ)
a. [NP PtcP NIII]		*!	*	
☞b. [NP PtcPIII NIII]			*	
c. [NP PtcP N]		*!		*

(14) Derivation of (2) - 3

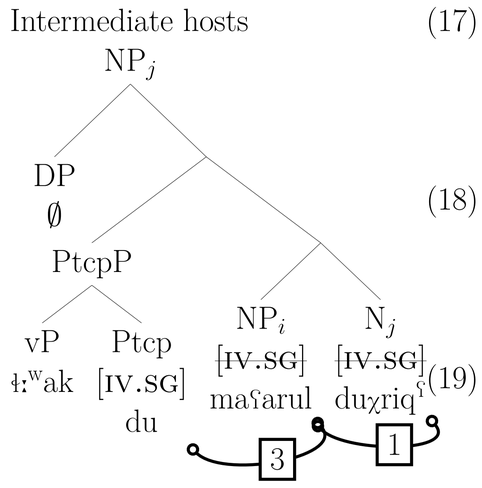
[NP NumP [PtcPIII N]]	*γ/D	LEFT(γ,NP)	*γ/N	MAX(γ)
a. [NP NumP [PtcPIII N]]		*!		
☞b. [NP NumPIII [PtcPIII N]]				

(15) Derivation of (2) - 4

[NP DemP [NumPIII [...]]]	*γ/D	LEFT(γ,NP)	*γ/N	MAX(γ)
a. [NP DemP [NumPIII [...]]]		*!		
☞b. [NP DemPIII [NumPIII [...]]]				

- Ranking *γ/NP below this constraint, allows for nominal adjectives to serve as intermediate hosts with their features deleted only after it has been copied to a higher position.

(16) Intermediate hosts



(17) Derivation of (3) - 1

[NP NP NIV]	*γ/D	LEFT(γ,NP)	*γ/N	MAX(γ)
a. [NP NP NIV]		*!		
☞ b. [NP NPIV NIV]			**	
c. [NP NP N]		*!		*

(18) Derivation of (3) - 2

[NP NPIV NIV]	*γ/D	LEFT(γ,NP)	*γ/N	MAX(γ)
a. [NP NPIV NIV]			**!	
☞ b. [NP NPIV N]			*	*
c. [NP NP NIV]		*!	*	*

(19) Derivation of (3) - 3

[NP PtcP [NPIV N]]	*γ/D	LEFT(γ,NP)	*γ/N	MAX(γ)
a. [NP PtcP [NPIV N]]		*!	*	
☞ b. [NP PtcPIV [NPIV N]]			*	

(20) Derivation of (3) - 4

[NP PtcPIV [NPIV N]]	*γ/D	LEFT(γ,NP)	*γ/N	MAX(γ)
a. [NP PtcPIV [NPIV N]]			*!	
☞ b. [NP PtcPIV [NP N]]				*
c. [NP PtcP [NPIV N]]			*!	*

6. Discussion: Alternative Approaches

- The operation-based account in Polinsky (2016) needs a non-local unrestricted operation and thus massively overgenerates. Copying is iterative and disregards barriers (Polinsky, 2016, 229-230).
- A global constraint based account has the sour grapes property and problems with gapped structures. Opaque interaction is not possible (McCarthy, 2009).
- In contrast, the present account employs a local, iterative copy operation, bounded by feature cooccurence constraint.
- It uses serial evaluation and counterbleeding interaction to solve the problems mentioned above.

References

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